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Crowdsourced Civic Issue Reporting And Resolution System

A PROJECT REPORT

Submitted by

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Under the guidance of,

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "Crowdsourced Civic Issue Reporting And Resolution System" is a bonafide work of "M SIDHARDHA (20221CSE0281), T SREEVARDHAN (20221CSE0282), P SASI KUMAR (20221CSE0259)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING, during 2025-26.

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named M.SIDHARDHA. T.SREE VARDHAN,P.SASI KUMAR, hereby declare that the project work titled **"**CROWDSOURCED CIVIC ISSUE REPORTING AND RESOLUTION SYSTEM**"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

Cities and rural areas often encounter civic issues (for example potholes, garbage buildup, leaking plumbing, malfunctioning streetlights, and drainage problems), which affect the quality of life of citizens in both cases. Traditional reporting mechanisms for civic issues have relied on manual complaints, protracted processes with little transparency and slow resolution times. Thus, for a number of reasons, many civic complaints go unresolved or are otherwise handled poorly. To resolve these problems, a Crowd sourced Civic Issue Reporting and Resolution System is being implemented. It is a technology-based solution that will help to increase citizen participation in local government and to resolve civic issues.Citizens can submit complaints using digital images, by entering descriptive information about the issue and, by using GPS coordinates to report the exact location of the issue. Once a citizen submits a complaint, the civic issue will automatically be categorized and forwarded to the appropriate municipal department. This process minimizes the amount of manual effort required to resolve civic issues by ensuring that all complaints are delivered to the relevant municipal department in a timely manner. Crowd sourcing has a large role in increasing transparency and accountability in order for users to view, vote and comment on problems submitted. this assists authority in their prioritizing when hitting issues based on urgency and impact on the public. higher levels of user engagement with a problem will enable authority to prioritize that engagement when distributing resources based on community supported participation, thus validating the legitimacy of the complaint submitted and decreasing duplicates from being submitted.

The Crowd sourced Civic Issue Reporting and Resolution System (CCIRRS) is a digital platform designed to enhance the efficiency and transparency of municipal (civic) complaint management. The system allows citizens to report local problems (e.g., potholes, overflowing garbage, water leaks, failed streetlights) using an app on either their mobile device or computer. Each complaint includes location data and supporting media in order that accurate identification of the problem can occur.The implemented CCIRRS provides a mechanism for real-time complaint submission, automatic categorization of all complaints, and direct routing of each complaint to the appropriate municipal department. The centralized administrative dashboard allows municipal authorities to efficiently monitor, assign, and track the resolution of issues. The inclusion of crowd sourcing features (e.g., up voting/collectively validating issues) ensures that the most serious issues are addressed based on the voice of the community.The results from the implementation of CCIRRS show a decrease in the time it takes for a municipality to respond to a citizens' complaint, an increase in accountability related to the resolution of complaints, and an increase in the transparency surrounding the delivery of municipal services. Citizens can track their complaints and receive notifications of progress. In addition, authorities can utilize data analytic tools to generate reports identifying the areas most frequently impacted by similar complaints, and allow authorities to mitigate future occurrences of similar complaints.

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Abbreviations

GPS	Global positioing system
HTML	Hyper text markup language
CSS	Cascading style sheets
JS	Java script
IDE	Integared development environment
API	Application programming interface
DBMS	Databse management system
CIRS	Civic issues repotyinf system
UI	User inetrface
UX	User experience
DFD	Data flow diagram
SQL	Structure query language
ER	Entity relationship diagram
RBAC	Roel bases access control
AWS	Amazon web services
GIS	Geographical informarion system

Chapter 1

Introduction

As cities and towns flood with people and grow, urbanization leads to increases of demand for civic authorities' services in these places, and citizens face many of their cities' basic civic issues every day (broken roads, overflowing garbage bins, broken water supply, broken street lights, open manholes, clogged drains). These problems create inconveniences for citizens but also pose real threats to their health, safety, and overall quality of life. Without solutions to civic issues and processes, sustainable development of cities cannot happen. The traditional methods of reporting civic issues such as submitting a complaint to a municipal office, writing a letter to complain, or calling in to report an issue, are slow, cumbersome, and inefficient. In addition, citizens have little or no confidence that their complaint has been resolved satisfactorily or that local government is dealing with it in a timely manner. Because citizens cannot easily check the status of their complaint, they become frustrated with the process, which leads to further erosion of their trust in local government. Because of the amount of time it takes to resolve complaints and due to poor communication (between the departments), many complaints go unresolved or lost, and the result is no resolution due to a lack of coordination and communication among municipal departments. Technological advances and the growth of mobile technology are creating more effective ways for citizens to document and report civic issues throughout their cities and towns. An online civic issue reporting and resolution system (Crowd sourced) allows residents to quickly easily communicate with the city's government officials about civic problems in the civic region or zone where they live. Residents can submit civic issues they encounter through a web or mobile application from anywhere without needing to meet with an official face-to-face. However, when reporting an issue via the app, residents are encouraged to upload images, descriptions and/or GPS locations that will positively and negatively affect accuracy of the report and resolving the issue will be streamlined due because of the detailed accuracy of the report.

1.1 Background

The aim of the project is to solve the problems in urban areas,villages,towns,roads and some where else with the help of our web site the public can upload the problem here ,for that we have created a web site with the help of visual studio code and java script,css,html,react.

Smart city programs and digital government systems serve as inspiration for this initiative to make cities be more efficient, more transparent, and focused on citizens. Using mobile applications, web-based systems, GPS, and cloud-based technology, there will be a way to report and manage the resolution of civic problems, in real-time.

A digital platform makes it easy for citizens to report their issues with supporting pictures and accurate location information. This makes it possible for agencies to have accurate data for many different problems and to respond quickly to those problems. Data collected from multiple civic users can also provide an opportunity for municipal agencies to determine areas that experience repeated problems, as well as the frequency in which they are reported. By doing so, municipal agencies can move beyond reacting to problems to proactively planning and developing preventive maintenance. The basis for this initiative, therefore, is the need to rethink how citizen-driven civic administration will work by utilizing technology to improve citizen participation.

1.2 Statistics

So according to the statistics report we have many category issues/problems like roads,water,drainage etc.. facing by public .

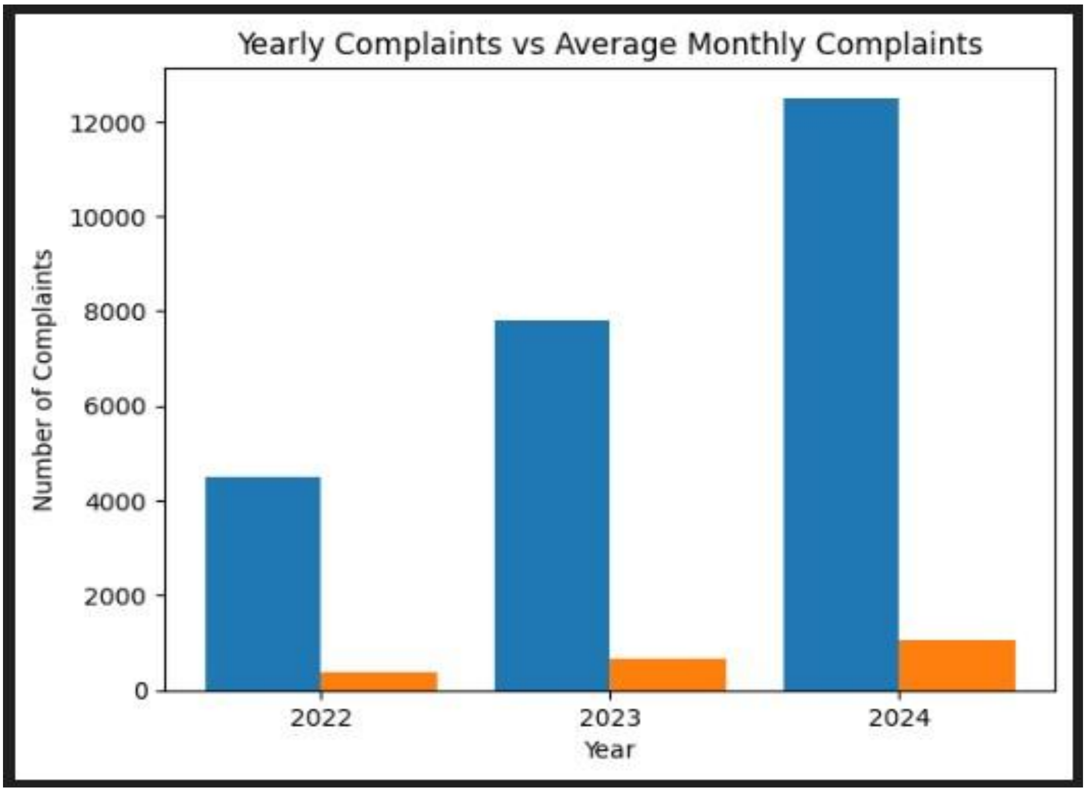


Fig 1.2.1 compilaints raised and sloved

From the above bar graph blue indicates pending and orange indicates solved before this web site

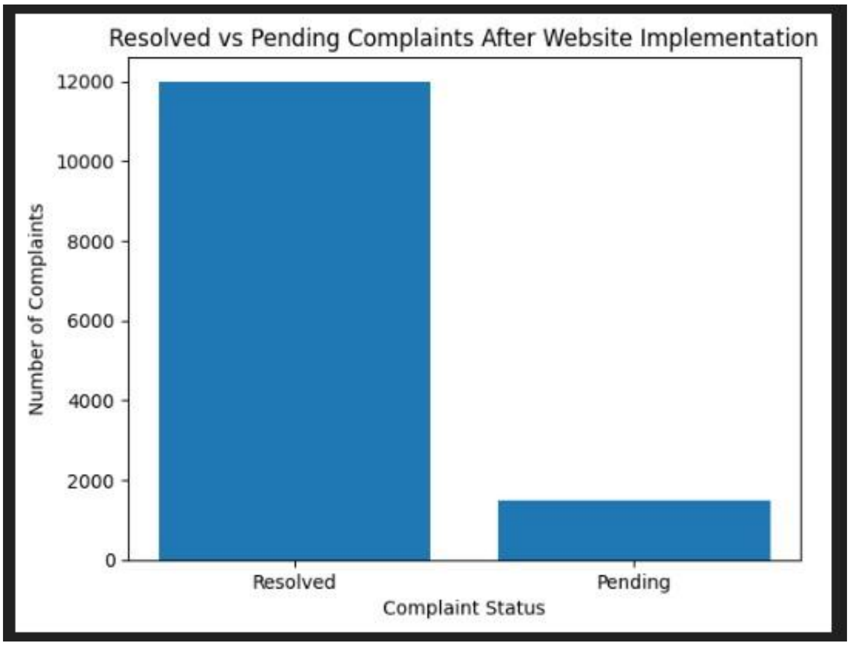


Fig 1.2.2 complaints that pending and resolved

No we can see that after creating this civic sourced web site the problems getting solved by the government

1.3 Proposed approach

The Crowdsourced Civic Issues Reporting and Resolution System has been designed to be both a web based and a mobile platform that will enable people to report civic issues directly and easily to the appropriate authorities in a structured and transparent manner. The system has been developed using a user-centered design within it, with the intent of being easy to access and to use regardless of age or previous experience.At the onset, users will have to register and log into the system using their own login credentials in order to establish an account. After successful login/authentication, users may submit complaints by simply selecting which category their issue relates to, such as damaged roadway, overflow of garbage, leaking water line, clogged drain or improperly functioning street light). In order to assist in correctly identifying the complained about issue by including images as well as the physical location of where the problem exists (via GPS integration or manual), the user will also need to provide location information.Once submitted, all data related to an individual user's complaint is automatically recorded into one centrally managed database with each complaint having its own unique complaint identification number. The complaint is then automatically routed to the appropriate municipal department according to both the specific problem category and the specific geolocation of the reported problem.

Each department is provided an administrative dashboard which allows for the management, assignment, and real-time monitoring of the status of the complaint being processed by that individual department.Users are able to track the status of their initiated complaint via an easy-to-navigate tracking feature located within their personal dashboard. Each time the status of a user's individual complaint changes (i.e.: when it is assigned to a municipal worker for action), the user will be notified via email and/or text message of the new complaint status.

1.3 Objectives

1. Create a Digital Platform That Citizens Can Use To Easily Report Civic Problems (via Web Or Mobile App)
2. Automate Routing Of Civic Complaints To Speed Up Citizen's Access To Civic Services
3. Improve Citizen's Access To Information About Complaints Made By Providing RealTime Updates On Complaints
4. Improve Communication Between Citizens And Local Governments By Creating A Centralized System For Reporting Civic Problems
5. Create An Organized System To Manage Complaints With Clear Categorization And Prioritization
6. Encourage Citizens To Participate In Governance By Using Crowd-sourcing Techniques Such As Up-voting And Providing Feedback
7. Create And Maintain An Efficient Centralized Complaint Database
8. Use GPS Technology To Identify Exact Locations Of Civic Problems
9. Provide An Administrative Dashboard So Local Authorities Can Track, Assign, And

Manage Complaints

10. Generate Analytic Reports And Visuals That Will Help To Identify The Cause Of Recurring Problems And Help To Determine Which Areas Are Most At-Risk For Civic Problems

11. Ensure That Data Is Handled Safely/securely By Using Authentication And Role-Based Access Control (RBAC/RBAC+)

12. Create Methods For Using Technology For Improved Civic Services To Contribute To A "Smart City.

1.4 SDGs

This project comes under the SDG 16 - peace justice and strong institutions

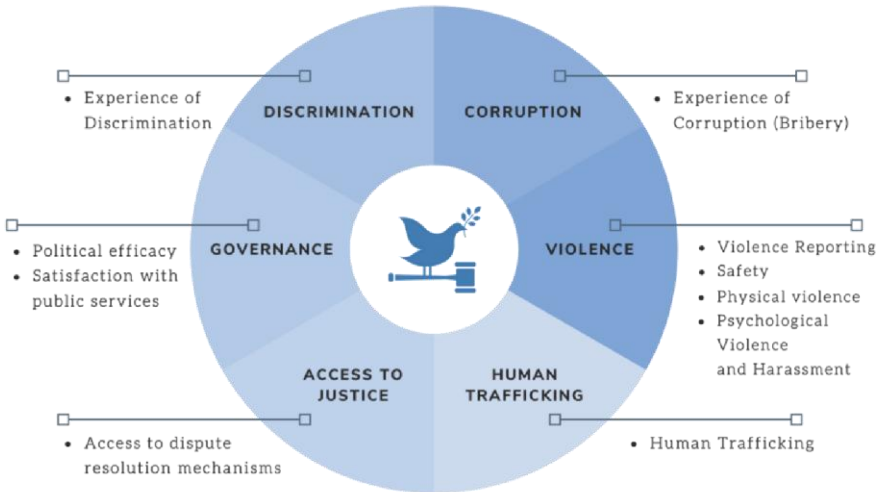


Fig 1.5.1 Sustainable development goals of SDG 16

1.5 Overview of project report

The Crowd-sourced Civic Issue Report & Resolution System is a web-based platform created to enhance the reporting and resolution of civic issues throughout urban and rural communities. The project's goal is to develop a transparent, efficient, and citizen-driven system that links citizens directly with local municipalities through digital technology.This project's focus will be to address issues most citizens face on a daily basis, such as potholes, overflowing garbage, leaking water, blocked drains, and failed streetlights. Traditional complaint systems are typically slow, poorly documented, and do not allow citizens to see the status of their complaints. To address these issues, the new proposed system will create a way for users to register their complaints online, as well as track their complaints throughout the process.The system is designed such that all users can register, log in, and submit an issue by selecting the category of problem they have, uploading appropriate photos, and providing details regarding the physical location of the issue.

There will be an administrative dashboard available for the municipal authority's use to record, monitor, and manage individual and overall complaints as efficiently as possible. The municipal staff will be able to see, assign; make updates, and close all complaint records in real time manner. At the same time as the municipal staff are doing so, the users will also be able to track the status of their complaints. Therefore, the proposed system will also provide users with notifications regarding the status of their complaints.

Chapter 2

Literature review

The number of people in cities and urbanized areas is rapidly increasing leading to many urban centers struggling with managing residents and service delivery. Governments at the local level are facing incredible challenges due to an increase in residents and the significant stress they are placing on the cities they reside in... In the last few years, several research projects and practical implementations have been completed that demonstrate the very real need for technology to be incorporated into civic reporting processes to make reporting and resolving civic issues more efficient. Khalil et al., (2014) suggest that traditional means of receiving complaints from citizens (in the form of paper-based forms, via telephone, and in-person) are near impossible to use effectively, as they do not allow for time sensitive reporting, are often delayed, and do not provide citizens with accountability and/or transparency regarding their claims. Consequently, the inefficiencies and ineffectiveness result in a general dissatisfaction of citizens and ultimately cause an erosion of trust in their local government. Based on this evidence, researchers have focused on creating technology delivery systems to enable/speed up the real-time submission of civic issues in order to facilitate communication between citizens and cities. Crowdsourcing has become one of the most researched approaches within this area and is defined, by Howe (2006).

Research on Civic Reporting Systems has progressed throughout time. The purpose of many studies has been to create and implement a civic reporting system. For example, Xiong et al. (2015) examined a mobile-based report submission platform that allowed citizens to submit geo-tagged reports of local issues, such as damaged physical infrastructure. Their results indicated that Geo-tagged reports led to quicker responses from city authorities than non-geotagged reports, since the city authorities were able to locate the issue more easily. Longitudinal studies have also examined how public feedback mechanisms such as upvoting, commenting and validation by the community affect the civic reporting systems that are in place. Chen et al. (2018) found that public engagement mechanisms have facilitated decision making for authorities regarding prioritization of issues based on the collective judgement of the public. In other words, if an issue receives more public support, the issue will generally be critical and receive resources from the authorities to resolve it compared to issues that received very little public support. Gupta and Kumar (2019) also indicated that the use of social validation decreases the number of instances of duplicate reports, and improves the quality of the data collected through clustering similar types of complaints. Civic reporting and GIS (Geographical Information Systems) have received increased attention as a developing field but have yet to reach their full potential for research in integration. The ability to visualize challenges has been greatly enhanced using GIS, providing an understanding of the locations of the challenges through clustering of events; this information can be utilized by cities to improve their future planning efforts. Li et al. (2017) published a study that illustrates how municipalities can use spatial analytics to enhance their decision-making processes. By employing spatial visualization tools, municipalities can determine where failures exist in their organizations; for example, if reports of potholes in a particular area are received frequently, it is likely that the infrastructure in the area may not be adequate and, therefore, will likely require further investigation.

Table 2.1 Summary of Literature reviews

S.no	Author and published year	methods	Key features	De merits
R. K sharma in 2019	Smart city complaint management system	Web based reporting with centrailized database	Imroved complaint tracking	No real time
A. Mehata in 2020	Crowd sourcing for urban governance	Mobile app with citizen participation	Public engagement in civic issues	Limited scalability
S.Gupta in 2018	Public service	Digital complaint portal with admin	Location based	No analytic
M.Khan and L.Verma in 2021	Civic issue monitoring	Integrated gps	Location based issue identification	High implementation

Chapter 3

Methodology

- Analysis of Requirements

Identified the issues with the current civic complaint system.

Gathered information from the citizens & municipal authorities regarding their needs.

Defined the function of system features like Complaint Registration, Tracking, Admin Dashboard, Notifications, etc.

- Designing the System

Designed the layout of the system (i.e., Frontend, Backend, & Database).

Created Data Flow Diagrams (DFDs), Use Case Diagrams, & Database Schemas.

Planned User Interfaces (UIs) that are easy to use and navigate.

- Developmental Phase

Developed the Frontend using HTML/CSS/JS.

Built the Backend logic utilizing Server Side Technologies (Python/Node.JS/Java).

Integrated Databases (My SQL/MongoDB), to save complaint information in the Database.

- Implementation of Crowdsourcing Mechanism

Created a mechanism for up-voting, and identifying duplicate issues.

Prioritized issues based on the number of reports and urgency.

Encouraged Community Participation in validating issues.

- Integration of GPS & Location Services

Integrated GPS feature to capture accurate geographic information for complaint location.

Saved the latitude and longitude of issues in the database.

Utilized maps to aid in the visualization of complaints.

Fig 3.1 system model type

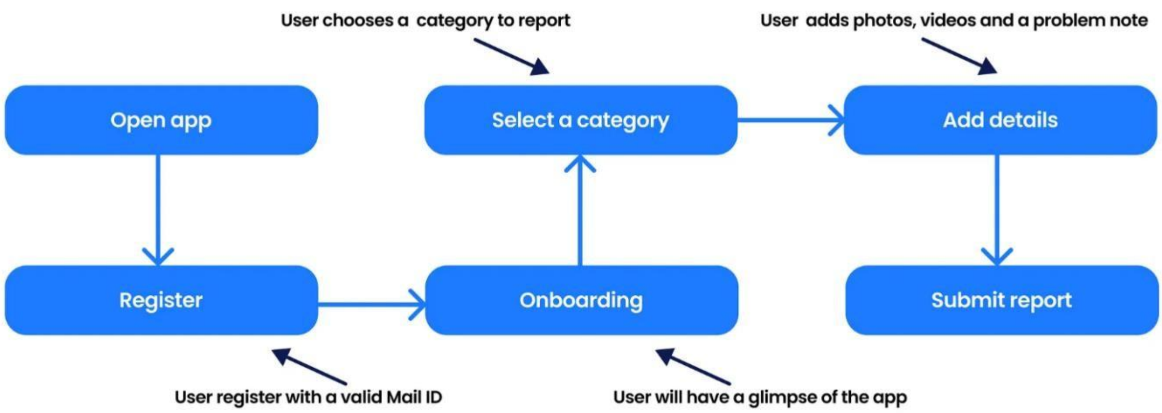


Fig 3.2 basic structure of this project

PROJECT MANAGEMENT METHODOLOGIES	PROS	CONS
Agile Methodology	• Flexibility and adaptability • Collaboration and teamwork • Continuous improvement and feedback	• Lack of predictability • Requires experienced team members • Can be difficult to manage for larger projects
Waterfall Methodology	• Well-defined requirements • Predictable and structured • Easy to manage for larger projects	• Limited flexibility • No room for changes or adjustments • Can be time-consuming and costly
Scrum Methodology	• Collaboration and teamwork • Flexibility and adaptability • Continuous improvement and feedback	• Requires experienced team members • Can be difficult to manage for larger projects • Lack of predictability
Kanban Methodology	• Visual management and continuous flow • Flexibility and adaptability • Focus on quality and customer satisfaction	• Limited structure and predictability • Requires experienced team members • Challenging to manage for larger projects
Lean Methodology	• Focus on efficiency and waste reduction • Continuous improvement and feedback • Structured and predictable	• Limited flexibility • Can be difficult to manage for larger projects • Requires experienced team members
Six Sigma Methodology	• Focus on quality and process improvement • Data-driven decision-making • Structured and predictable	• Can be time-consuming and costly • Requires experienced team members • Limited flexibility
PRINCE2 Methodology	• Structured and predictable • Focus on risk management and quality • Suitable for complex projects	• Time-consuming and expensive • Requires experienced team members • Flexibility constraints Source: CRM.org

Fig 3.3 types of methodologies

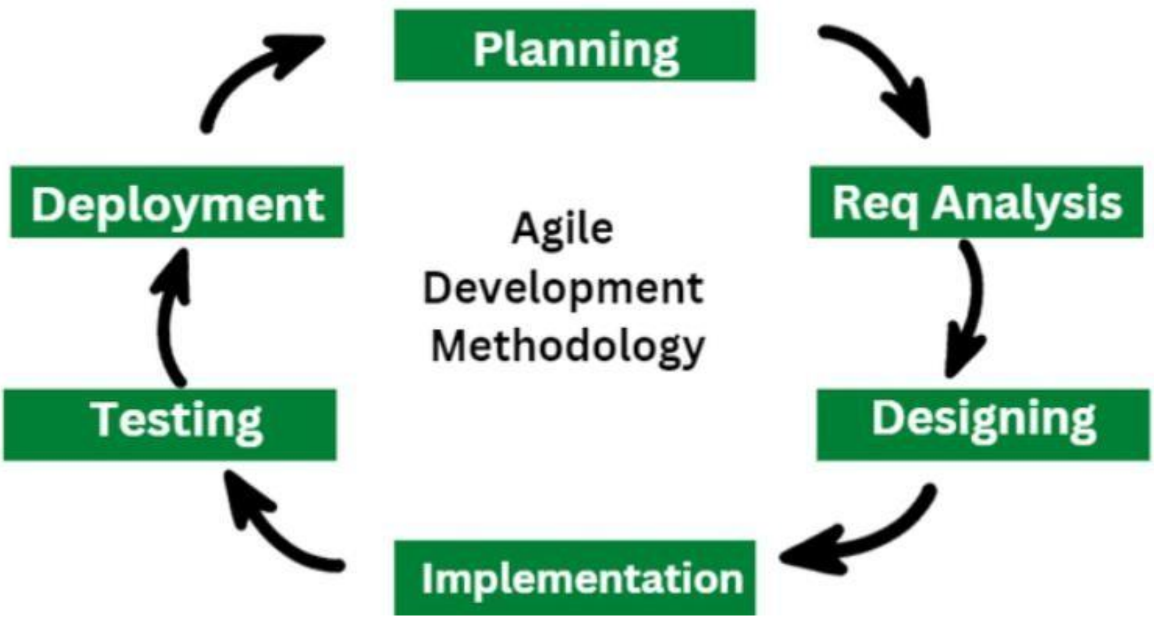


Fig 3.4 structure of agile

In this project we will use agile methodology

Agile methodology is used for model development,continuous updates and user feed back integration The Crowdsourced Civic Issue Reporting And Resolution System follows an agile development methodology to allow for flexible design, continuous improvement, and modular development. Since the ultimate goal of this project is to be citizen-focused, and will depend heavily on citizens to interact with the system, it is expected that a lot of the requirements may change as feedback comes from both citizens and other entities, The agile process allows for the project to be developed in small steps through iterations or sprints in which an individual feature (or module) can be built, tested and improved on, which minimizes risk, provides earlier detection of errors, and gets completed working features to the end user faster. In addition, the agile process supports continual testing and refinement of the product, which ultimately leads to an increase in both quality and performance of the overall civic issue system. The agile process will help ensure that the continued development and delivery of the system through regular updates, new features and scalability, the process aligns with the ever-changing dynamics of the real world and continues to meet the needs of citizens.

Chapter 4 Project Management

4.1 Project timeline

Gantt Chart Shows

- Project Phases
  - ➢ Requirement Analysis, Design, Development, Testing, and Deployment phases are easily identifiable using the Gantt chart.
  - ➢ This view of the Gantt chart allows for an easier way to break down the project into different stages for better project management and understanding of workflow. ● Start Time of Each Task
  - ➢ The Gantt chart provides a way to determine exactly when each stage of the project starts (week/month).
  - ➢ This enables a project to have the proper scheduled task/on time commencement of task based on the project schedule.
- End Time of Each Task
  - ➢ The Gantt chart provides a way to see roughly when each task will be completed.
  - ➢ This aids in avoiding delays in the project and provides enough time for a project to meet deadlines. ● Task Duration
  - ➢ The Gantt chart provides the amount of time the task will require to complete as represented by the length of the horizontal bar.
  - ➢ This assists in estimating the total amount of work for each task as well as determining the best use of resources available. ● Tasks that Overlap
  - ➢ The Gantt chart provides a way of visualising tasks that occur concurrently (ie; development and testing).
  - ➢ By performing more than one task at a time, this will allow for a reduction in total project length and increasing the overall efficiency.
- Project Progress Tracking
  - ➢ The Gantt chart provides a means to evaluate project progress against the planned timeframe.
  - ➢ This allows for early detection of project delays and enables corrective action.

1. Project Development

   - Project Development requires all planning to be completed prior to the initiation of any work.
   - This means that each phase of development (Requirements Analysis, Design, Development, Testing and Deployment) should have definitive timeframes.

2. Scheduling Tasks

   - Scheduling is the method used to determine when work will be started and completed.
   - Scheduling will assign tasks logically and in a clear manner.

3. allocating Resources

   - Resource allocation is the method to assign team members to specific tasks and to ensure that team members are not overworked and/or underutilized

4. Monitoring Progress

   - Monitoring progress is done by comparing the actual progress to the planned schedule to determine if there are any delays for a task, and if there are, it will allow for timely corrective actions to be taken.

5. Managing Dependencies

   - It is important to recognize that some tasks will be dependent on the completion of other tasks. (ex. Testing cannot be done until Development is complete.)
   - The use of a Gantt chart will allow for the recognition of these dependencies.

6. Meeting deadlines for a project

   - The purpose of project management is to deliver all project tasks within the planned timeframes so that the project can be completed on time.

To prepare Gantt chart Use open source tools GanttProject or Google Sheets. It can also be prepared using MS Excel, or MS Project that may require license.

Project Planning

Table 4.1 Project planning timeline

Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Project initiation (i)	■														
Selection of topic	■	■	■												
Background (ii)		■	■												
Objectives (iii)		■	■	■											
Methodology (iv)			■	■											
Proposal			■	■	■										
Literature review (v)				■	■	■	■								
Design and Ananlysis					■	■	■	■							
System Requirement Phase (vi)					■	■	■	■							
System design phase (vii)						■	■	■	■						
Functional unit design phase (viii)							■	■	■	■					
Report								■	■	■	■				
Final report											■	■	■		

(i) Project initiation - Live Projects, Projects of national importance (Smart - Environment, Mobility, Governance, Building and living, People, Economy, Renewal energy, Water conservation, Waste management, Health, Education, Tourism, Irrigation, Cities), Area for projects (Communication, Embedded systems, Signal and Image processing, VLSI, Controls, Networking, Security and cryptography)
(ii) Background - Background, approach, expected results
(iii) Objectives - Statements that describe the elements to achieve project aim. Writing an objective that is SMART - Specific, Measurable, Attainable/Achievable, Realistic, and Time-bound.
(iv) Methodology - Enlist and briefly describe the different methodology. Briefly describe each stage of the applied methodology , but discuss in details relating the various stages to implement the project.
(v) Literature review - Include a brief description with appropriate illustrations. Discuss the concepts, approach, methods, analysis, and issues adopted in part or full of your approach. Identify inconsistencies, gaps and contradictions, differences. Suggest improvements
(vi) System Requirement Phase - Datasheets, Identifying initial conditions, Identifying input parameters, Identifying system outcomes, Identifying relations, Identifying system constraints
(vii) System design phase - determining functional blocks, Identifying process flow, Identifying inconsistencies, Identifying interfaces, System design and analysis, developing a integrated test plan
(viii) Functional unit design phase - Identifying components, component datasheets, compare components, Unit design and analysis, developing a unit test plan

The Gantt chart details the stages of a 15-week project schedule, clearly defining all project phases from initiation to submission phase. The initiation phase starts with identifying and selecting a topic, establishing the project's scope. The following two phases address the background study needed to understand the problem adequately, defining specific SMART objectives that specifically identify the project's goals and/or analyzing existing systems to identify any gaps in existing research as well as creating an adequate theoretical framework for the project itself. After completing the first two phases, the third phase of the project focuses on determining the proposed project methodology; in other words, how to implement the design and develop the system. The next two phases are completed during the first two weeks of the project; the researcher will be conducting a literature review and identifying different systems as well as performing an analytically assessment of relevant literature and solidifying the proposal for submission for approval. The next three phases will involve designing and analyzing the system; this phase includes the system requirements analysis, the system design, the prototyping of the system before implementation, and initial development of the system (build of the system).

4.2 Risk analysis

Implementing the Crowdsourced Civic Issue Reporting and Resolution System will carry numerous risks that need to be identified, analyzed, and managed for successful implementation. The two major risks are: Technical risks which include software bugs, server outages, and database problems. Furthermore, With the system being reliant on the Internet users submitting and tracking complaints in real-time may be hindered by a poor network connection or internet quality. Another significant risk is data security and privacy; as this platform will collect personal identifiable information (PII) about users and their locations, if adequate security measures are not in place, this information could be accessed and exploited for malicious purposes. In addition, duplicating or misleading complaints submitted by users who are attempting to submit the same, irrelevant, or false complaint will cause issues with system overload, as well as decreasing system reliability.

The next risk is with user adoption. Due to a lack of knowledge or understanding of the platform, citizens may not actively engage with the platform. Current engagement level determines how much social impact the system will produce. Performance and scalability risks are also important to consider; by increasing the number of users rapidly, the system needs to maintain its overall functionality. As more data is stored in the system, it needs to be able to perform efficiently and effectively. Delays from local or state government in resolving reported complaints decreases users' trust in the platform. As well,

Fig 4.1 example of PESTEL analysis

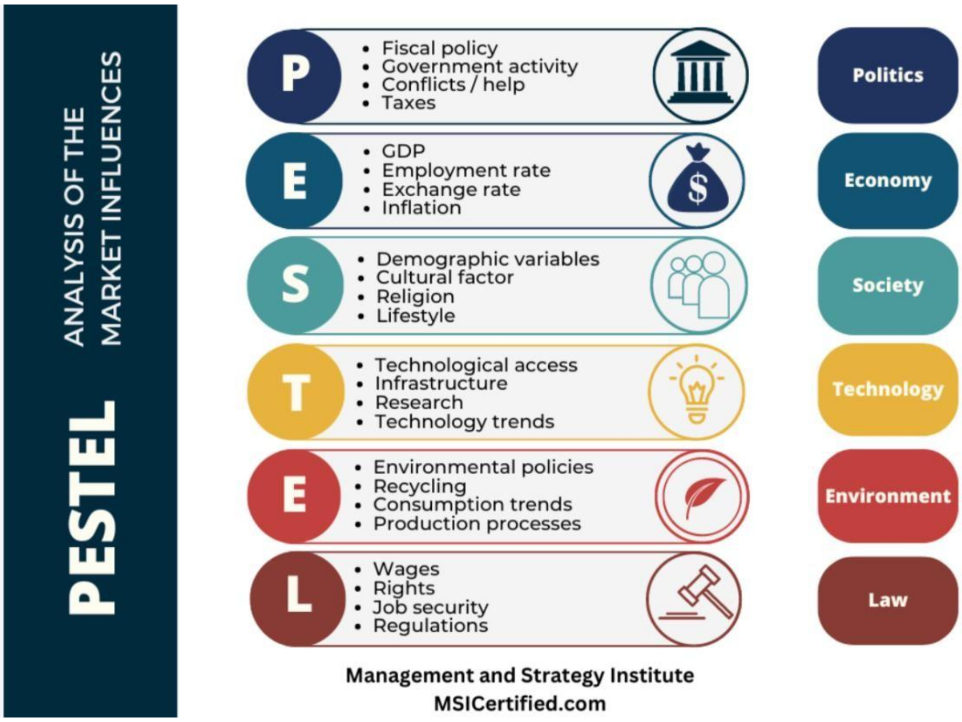


Fig 4.2 PESTEL analysis

4.3 Project budget

Fig 4.3 example model of project budget

Chapter 5 Analysis and Design

5.1 Analysis

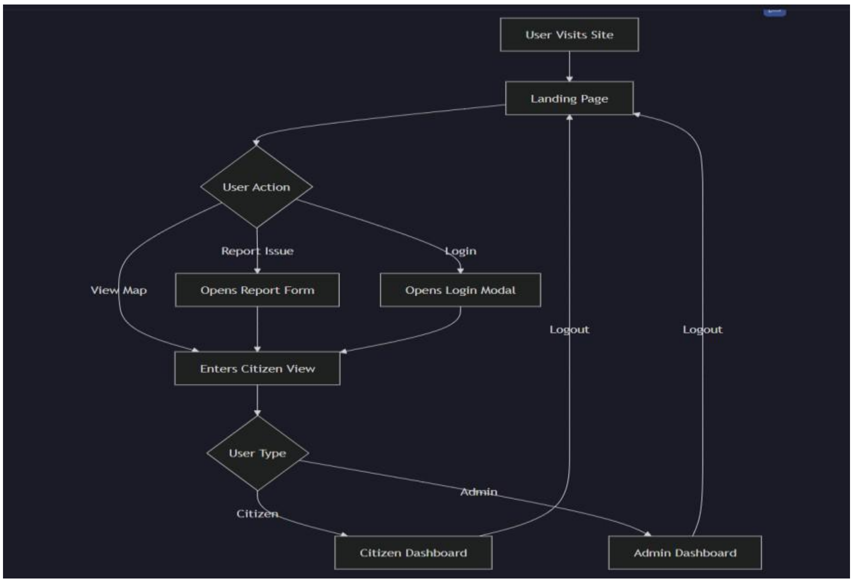
The Crowdsourced Civic Issue Reporting and Resolution System's Analysis phase is intended to provide an understanding of problem areas, user requirements, and system capabilities. The primary goal of this system is to provide an avenue for individuals to report their civic issues (potholes, garbage buildup, etc.), and have them addressed in a timely manner by the appropriate municipal authority. During the Requirements Analysis phase; both functional and non-functional requirements will be established for the system. Functional requirements will include (but are not limited to) user registration and login, reporting an issue with images and GPS coordinates, tracking live complaints, providing notification of updated complaint status, and providing municipal authorities who manage and resolve complaints with an administrative dashboard. Non-functional requirements will include system scalability, security, usability, and performance. The system should be able to process multiple users at once while maintaining data privacy and delivering a fast response time. Stakeholder analysis is a key component during the requirements phase. Key stakeholders include citizens (end users), municipal authorities (administrators), and system administrators. Citizens will require an easy-to-use, accessible website to report their issues; while municipal authorities will require efficient tools to monitor, manage, and resolve any complaint. Additional data analysis will also provide insight into the most frequently reported issues, as well as the most dangerous areas of the municipality, which will serve to enhance overall urban planning. The system will undergo a feasibility analysis to ensure that it can feasibly be developed, and the results of the technical feasibility study will provide reassurance to those involved that the project can be successfully delivered.

5.2 Requirement analysis

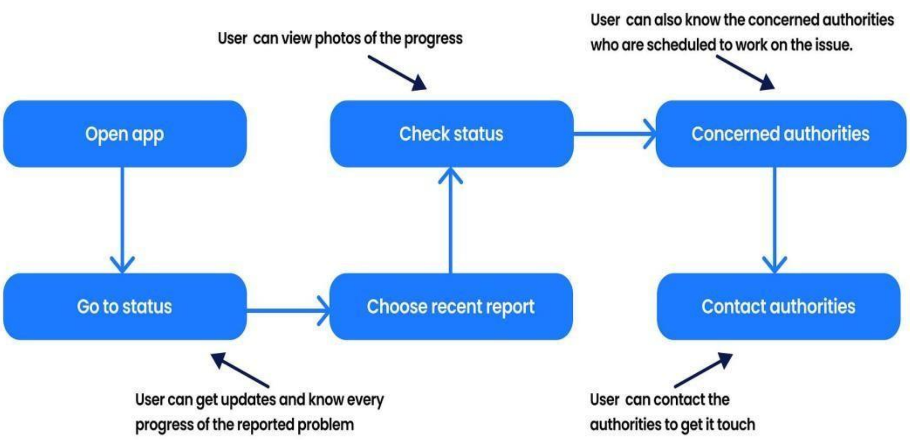
The process of analyzing requirements is one of the most important parts of building “Civics Issues Report and Solution System (CIRS)” as it will help us determine how the CIRS system will be built correctly by defining and documenting requirements from end users and stakeholders (users and administrators) for the CIRS system to meet the needs of users and deliver successful implementations. Citizens will be the main users of the CIRS system. The system will be used by citizens to report civic issues (e.g. potholes, litter, water leaks, defective streetlights). Therefore, Citizens will want a simple system that allows them to report common civic issues by attaching an image or description of the issue, with the ability to "tag" or provide the GPS location where the issue was reported. Citizens will also want to track the progress of their complaint and receive notification of the status and resolution of their complaint. Municipal Administrators will be responsible for administering the CIRS system for their Municipality, which includes managing the resolution of the civic issue(s) reported by Citizens. To be able to resolve the civic issues reported, Municipal Administrators will need access to a dashboard that will allow them to view all registered citizen complaints and classify the complaints by priority, assign work to Field Workers, and update the status of each complaint. In addition to being able to view complaint data for each citizen, Municipal Administrators will also need access to analytic data and reporting to provide insight regarding trends, issues, and assist in the decision-making process.

5.3 System design

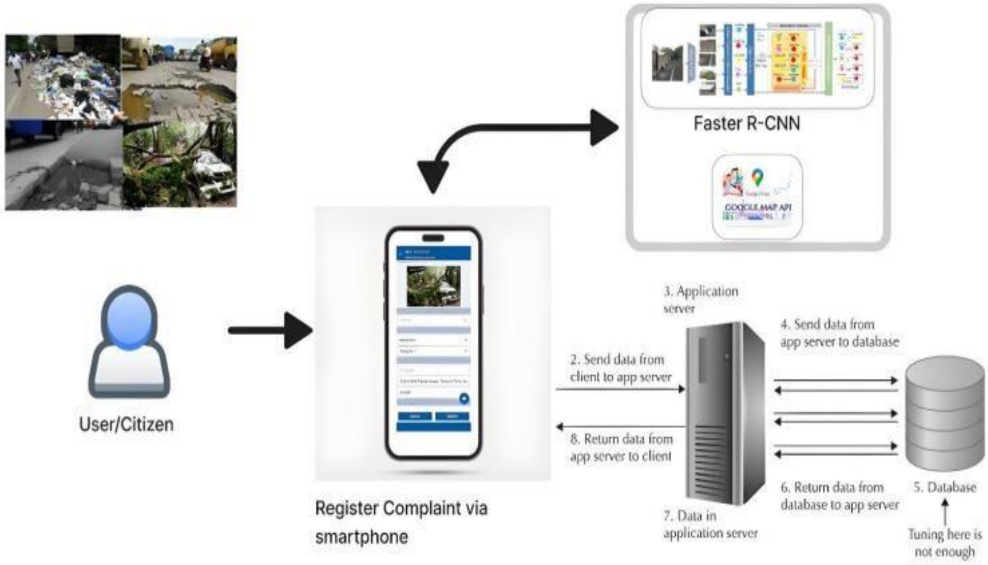
During the Design phase, we take the requirements that we gathered and analyzed from our previous phase and create a structured system design that will be used to develop/build the system. This system will be built on a multi-tier architecture consisting of a Presentation Layer (User Interface), Application Layer (Business Logic), and Data Layer (Data Storage). The Presentation Layer will be designed in such a way as to allow users to easily report issues they are experiencing by using user-contributed images, adding descriptive text to their reports, and tagging locations to their reports using GPS technology. The system architecture consists of a number of modular components, such as user management, issue reporting, issue tracking, notification system, and administrator dashboard. The User Management module will manage how users log into the system and how users change their profile information. The Issue Reporting module allows users to submit complaints, while the Issue Tracking module allows users to check on the status of their reported complaints. The administrator dashboard consists of functionality that allows the administrator to view the complaints submitted by users, and then to prioritize the complaints for addressing, assign the complaint to users of the administrator's staff for resolution, and keep the records of the resolution up to date. An important component of our design is the database design. The tables within the database will store records of users, issues, locations, and status updates. The relationships between the tables will facilitate data storage and retrieval operations. Our solution architecture may include the use of cloud-based databases as a method of providing maximum scalability and reliability for our use case. APIs will be used as the method of interfacing between the frontend and backend of the application. We will also implement security features for the protection of user data, such as user authentication, user authorization, and data encryption. Our solution architecture may include a Notification Design that will support various notification methods (SMS, e-mail, etc.).



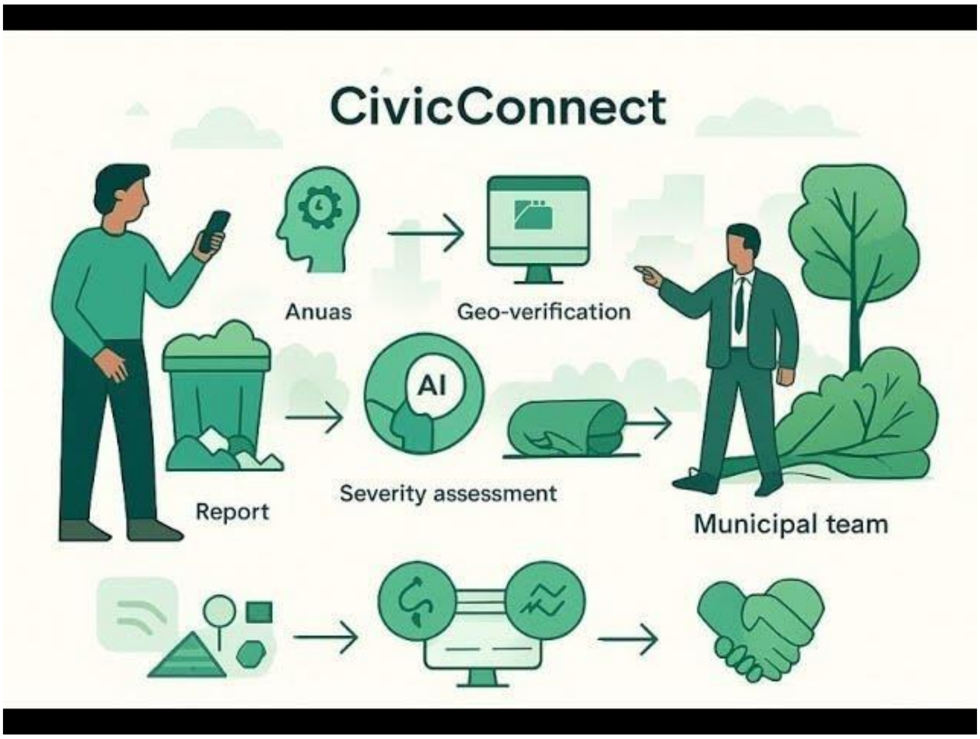
5.3.1 system architecture diagram



5.3.2 flow chart of the project



5.3.3 refrence picture of this project



5.3.4 simplest from this project

Chapter 6

Hardware, Software and Simulation

6.1 Hard ware

6.1.1 Citizen Device Hardware (User side)

Citizen's mobile phone/personal computer is the primary hardware that make up this system. The use of smartphones in the system is especially important because they have built-in sensors (e.g., camera and GPS Module). The users will take pictures of a problem such as potholes, litter, or damaged infrastructure (with the camera), providing physical evidence the authorities need to determine if an improvement is warranted. GPS is also used to tag the location and provide accurate mapping showing where the problem is located within a geographic area. Modern smartphones also come equipped with enough processing power, memory, and internet connectivity (Wi-Fi or mobile data) to allow users to easily upload images or a description, and/or location details when using the system. The convenience of a touchscreen on a smartphone user interface further increases usability for all community members, including non-technical users. Personal computers (PC) or laptops will also be used by individuals to access an online version of the system; however, those users may have to enter a location manually (as opposed to the GPS capabilities of a smartphone).

6.1.2 Municipal Authority Systems

Municipal authority administrators utilize laptops and desktop computers to manage monitor reported issues utilizing high-performance computer systems that feature moderate to high levels of processing capability. Thus, these systems contain sufficient RAM/storage/processing speed to accommodate numerous applications for Dashboard displays, Analytical Tools, and Supporting Largescale Data Sets.

Typical components of An Administrative System consist of:

- -Reliable Display(s) including high Definition Quality providing improved map/report imagery.
- -Input Devices (i.e., Keyboard, Mouse) allowing the management/execution and storage of data gathered from inputs.
- -Reliable Network Connectivity (e.g., Ethernet or Wi-Fi) that provide uninterrupted communication.

6.1.3 server hardware

Server Hardware is basically where everything starts and the heart of the system. It is the physical server or the virtual server on the cloud where the application is hosted, user requests are processed, and data is stored.

Server hardware consists of:

- High-performance CPUs (Processors) that will be able to process user request simultaneously.
- Large amounts of RAM for quicker execution of all backend processes.
- Storage devices (HDD or SSD) where all user data, imaging, and issues can be stored.

Backup systems that will prevent the loss of said data

Modern implementations rely heavily on cloud platforms (e.g. AWS, Azure, or Google Cloud) rather than physical servers. As these platforms are designed to offer multiple users at the same time by offering scalable virtual hardware, they can effectively handle larger amounts of user requests without impacting performance.

6.2 Software

s1: Integrated Development Environments

Integrated Development Environments provide software developers with an all-in-one environment in which to create, write, edit and debug their code. There are a number of wellknown IDEs available to developers, including Visual Studio Code, Eclipse and PyCharm, each offering similar core capabilities such as syntax highlighting, auto-complete and debugging, as well as extensions which allow developers to code in different programming languages. For this particular project, we have elected to use Visual Studio Code as our primary IDE due to its light weight and its ability to support front-end and back-end web development.

2: Version Control Systems

Version control systems allow software developers to track the history of their code and collaborate with multiple software developers to update the code. Git, which is used in conjunction with tools such as GitHub or GitLab, is the most widely used version control system.Version control systems are valuable tools for:

- Tracking code history
- Supporting collaboration with multiple developers
- Reverting back to a previous version of the code if an error occurs

3: Frontend Development Tools

Frontend development tools are the tools used to design and develop a system's User Interface (UI). Examples of the types of tools that are considered to be frontend development tools include:

- Web-based editors for HTML, CSS and JavaScript
- Frameworks for developing UIs such as React and Angular
- UI libraries such as Bootstrap

Chapter 7 Evaluation and Results

7.1 Evaluation

All evaluations have been completed following industry best practice methodologies in order to comply with acceptance/rejection criteria. All evaluation methodologies include both qualitative and quantitative measurements. Evaluations of systems must be performed based upon multiple sources of data in order to provide complete database for analysis of results. For example: Evaluation/Testing will include both formal and informal tests using a number of different testing techniques and tools to provide data to the evaluators. Evaluations will be completed within 90 days, and evaluation reports will provide overall results to both users and municipal agencies. Evaluation reports for the Crowdsourced Civic Issue Reporting . In conclusion, the Crowdsourced Civic Issue Reporting and Resolution System will be evaluated over a period of 12-18 months following implementation in order to assess how well it meets both user needs and achieves intended outcomes. Usability testing examines the degree of simplicity and ease of use associated with a system from a user or customer perspective, which includes both citizens and municipal authorities. The system's user interface must be uncomplicated, easy to use, and accessible to all users, regardless of their level of technical expertise. While performing usability testing, users provide valuable feedback, which can be utilized to identify barriers to navigation and use within the system. This continuous evaluation serves to better the usability of the system.

Functional testing determines whether the system provides working, integrated functionality (such as user registration, reporting an issue, digital photos, GPS location, and administrator access) can be performed for all users in an efficient manner. The accuracy and reliability of the system are critical evaluation criteria. An accurate and reliable system maintains correct data regarding issues, locations, and statuses on a consistent basis.Evaluation of security is also important because user data must be protected against unauthorized users. The implementation of proper user authentication and encryption mechanisms is required for determining the overall security of the system.Scalability is another criteria considered as part of the evaluation process. Scalability measures how well the system will be able to store and retrieve data for an increasing number of users, and whether or not the performance of the application can be affected by an increase in users and/or data. Positive user feedback demonstrates that the system meets its intended purpose and is easy to use; assessment of the impact will show specific examples of benefits made possible by the use of the system, such as faster resolution of reported issues, improved communication between users and authorities, or an increased level of active participation by the public, among others. Overall, conducting these evaluations assures that the system will provide an effective and efficient mechanism for supplying citizens and municipal governments with a viable solution.

7.2 RESULTS

The implementation of the Crowdsourced Civic Issue Reporting and Resolution System delivered effective and practical results in the overall improvement of reporting and resolving civic issues. The Crowdsourced Civic Issue Reporting and Resolution System enabled users to register, login and report issues such as potholes, garbage accumulation and water leakage complete with images and location information. All reported issues were saved in the database and visible to administrators in real-time, providing transparency and efficiently tracking issues.The admin module performed as expected, allowing administrators to view, categorize, assign and update the statuses of reported issues. Users received real-time notifications for status tracking of their reported complaints to facilitate improved communications between citizens and municipal authorities, and to increase citizen trust and engagement in the system.Performance results indicate that the system successfully handled multiple requests concurrently without undue delay. The interface was considered to be user-friendly, enabling even those without technical backgrounds to report civic issues with ease. The integration of location tracking enabled accurate identification of problem areas thereby improving response time and resource allocation.

Finally, the system evidenced reliability by upholding consistency and security of data resulting in the safe disposal of user data. Additionally, the structured

Chapter 8 Social, Legal, Ethical, Sustainability and Safety aspects

The Crowdsourced Civic Issue Reporting and Resolution System has a wide range of social, legal, ethical, sustainability, and safety ramifications that arise from its direct interaction with the public and local (government) authorities. It fosters social interaction through an active citizen participation mechanism that allows users to contribute to community development and civic issues. The system improves the relationship between citizens and city authorities, provides greater transparency of local government agencies, and encourages people to take collective responsibility for maintaining the infrastructure. This leads to more improved living conditions and a more responsive civic administration.

The legal ramifications of the system include the user data being appropriately managed in accordance with the data protection and privacy legislation. User data (i.e., name, contact number, location) must be collected only after obtaining user consent, and this data will only be used for the purposes it was initially collected for. The operation of the reporting system must also comply with all applicable laws and regulations concerning the provision of digital services and the management of public complaints. The system must also ensure that the terms and conditions it establishes are clear enough to allow for greater accountability on the part of the provider and to prevent legal actions from occurring.

The ethical ramifications of the system include the ethical treatment of all persons using the system; ensuring the overall accuracy of the information that is provided; preventing any misuse of the system, including false reporting, spamming, and/or posting information that is misleading; and providing for the maintenance of confidentiality concerning all individual users, where appropriate, so that the individual's identity cannot be disclosed.

8.1.1 Social aspects

- Encourages community involvement with local government
- Promotes interaction between citizens and the government
- Improving the level of accountability in Governance
- Encourages communities to take ownership of their neighborhoods ➢ Improving the overall livability of the cities.

8.1.2 Legal aspects

- Data Privacy must be in place
- Collection of personal information requires an Individual to give permission (consent)
- Public access will be provided in terms of the use of information
- Government Digital Services Act must be followed
- Have accountability for the way we use and store data.

8.1.3 Ethical aspects

- Preventing false reports and misuse of the reporting system
- Fairness to All Complainants
- Confidentiality and anonymity for Public
- No bias on issue priority
- Honest and responsible use of the reporting system

8.1.4 Sustainability aspects

- Encourages effective resource management (waste, water, roads)
- Assists in long range sustainable urban development
- Reduces the use of manual paper through electronic means
- Relies on cloud based technologies to make efficient use of energy
- Provides a means to elevate the level of environmental consciousness

8.1.5 Safety aspects

- Allows for rapid reporting of hazards (potholes, open drains, etc.)
- Enhances the safety of citizens and the maintenance of infrastructure
- Provides secure authentication of users and protection of their data
- Provides protection from cyber threats and unauthorized access
- Maintains the reliability and availability of the systems involved

Chapter 9 Conclusion

Make sure that the system enables users to access a fully working and legal application; as users become more accustomed to using the app, they will have developed new behaviours which will ultimately result in more effective reporting methods and faster resolution times.

Using a crowdsourced methodology to identify issues will provide municipalities with further insight into the state of civic infrastructure through the use of citizen reporting (potholes) combined with GPS-tagged infrastructure recorded in the app. The use of crowdsourced methods will also make it easier for the app to gather large quantities of data in a short time period, resulting in an open data repository that all municipal agencies can use when assessing their public infrastructures.

In summary, the Crowdsourced Civic Issue Reporting and Resolution System provides an effective method of facilitating communication between citizens and municipal authorities, helping municipalities to manage civic issues more efficiently and providing a transparent method of resolving civic issues throughout the community. The Crowdsourced Civic Issue Reporting and Resolution System will demonstrate to municipal authorities how technology can provide a centralised and efficient method of reporting civic issues.

Crowdsourced Civic Issue Reporting and Resolution System is moving away from the usual way of doing things, which is reactionary to a new way of providing government service based on technology, hence, a different type of engagement for citizens participating in their own urban community's development through the use of technology providing the ability to connect with their government service providers. In addition to helping citizens easily report civic issues, this project will create an ongoing and systematic method for recording, prioritizing, and resolving those issues with more efficiencies and accountability.

One significant benefit of this new way to manage civic data is that it creates a central and structured database that allows governments to analyze this information to determine the highest number of complaints and where they are located, as well as looking for service inefficiencies, allowing government agencies to implement a preventative model of service vs.

just a correction model. Thus, this move away from complaint management will make this tool a decision support system for future urban planning and policy development.

The other important thing about this whole system is that it is proof that you can use cloud computing, GPS-based location technology, and real-time notification in order to create a responsive and scalable solution. Another benefit of this system being modular in design is that it will allow for easy modifications, enhancements, or upgrades as new technologies become available, such as those mentioned above.

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Base Paper

The paper "Crowd Sourced Civic Issues? Analyzing the Extent of Socio-Demographic Inequality in Crowdsourced Civic Participation on FixMyStreet Brussels" studies how citizens use online civic reporting platforms. It focuses on the FixMyStreet system, where users report issues like potholes, garbage, and infrastructure problems. The authors analyze more than 30,000 reports collected from Brussels. The study reveals that participation in such platforms is not evenly distributed among all citizens. Higher-income and digitally literate groups tend to use the system more frequently. In contrast, lower-income and minority communities are underrepresented. This creates a participation gap in civic engagement. The paper also highlights that access to technology and awareness significantly affect reporting behavior. It suggests that civic platforms must be more inclusive and accessible. The findings emphasize the importance of designing systems that encourage equal participation. This research supports the need for improved civic reporting systems like the proposed project.

Title: FixMyStreet Brussels: Sociodemographic Inequality in Crowdsourced Civic Participation.

Authors: Burak Pak, Alvin Chua, Andrew Vande Moere.

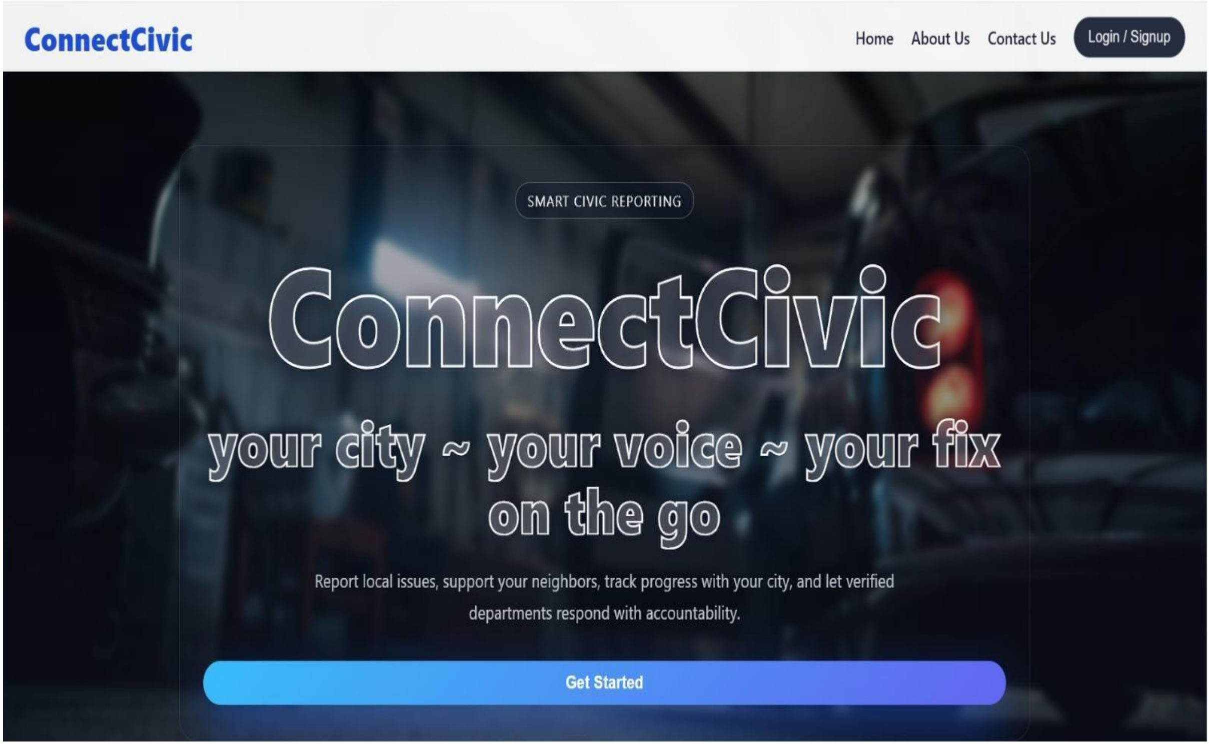
Published In: Journal of Urban Technology, 2017

Link:https://www.researchgate.net/publication/311203042_Who_is_Really_Using_FixMyStreet_Analyzing_the_extent_of_Socio-Demographic_Inequality_in_Crowdsourced_Civic_Participation_on_FixMyStreet_Brussels

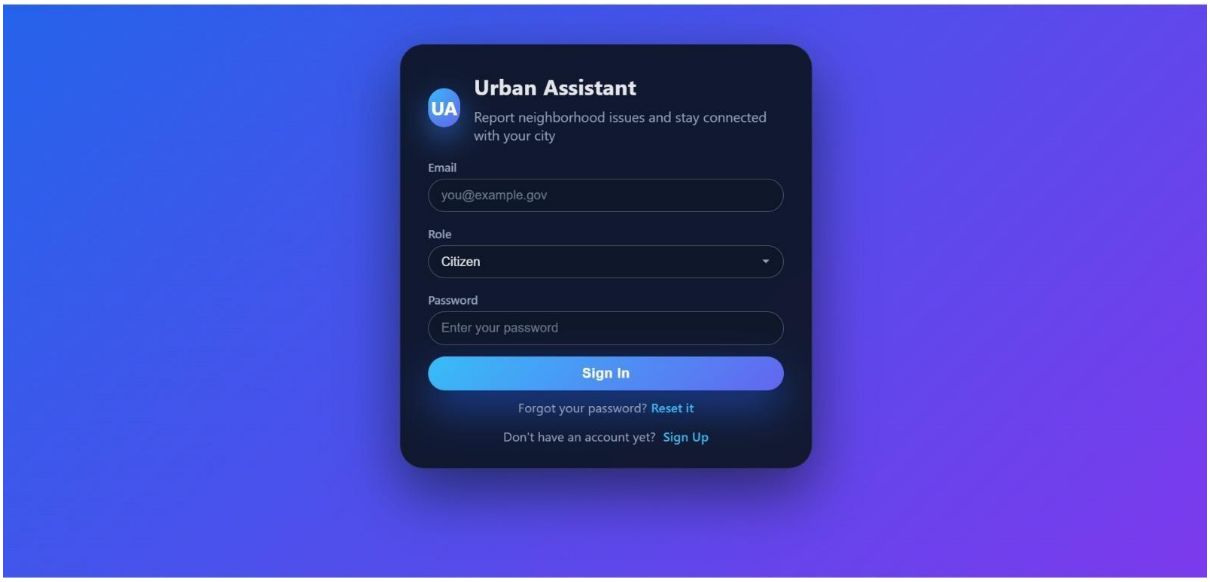
Appendix

The appendices include supporting material that enhances understanding but is too detailed for inclusion in the main text

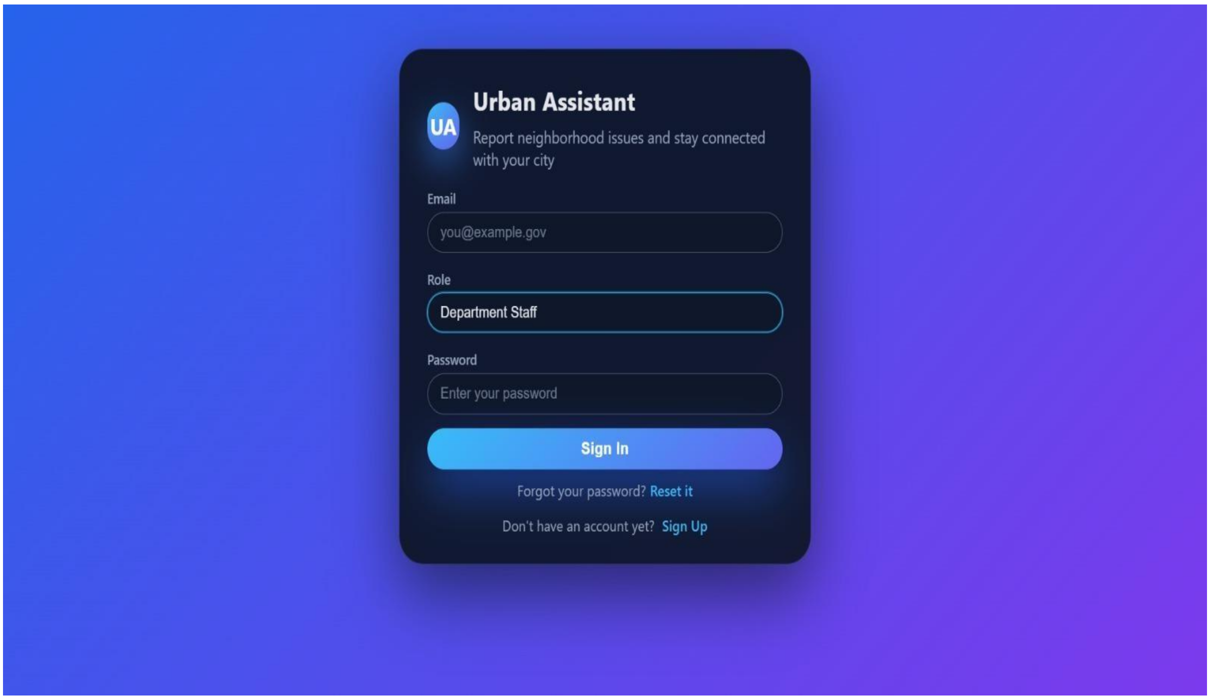
Appendix A – Screenshots and System Interfaces



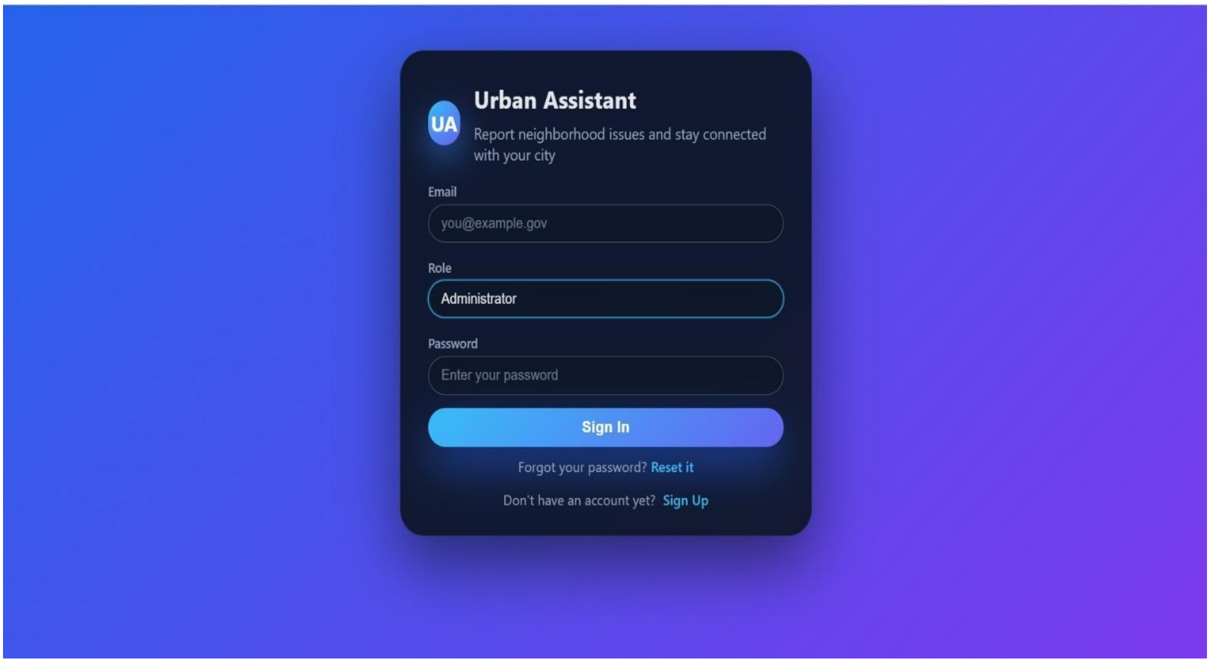
A1 cover page



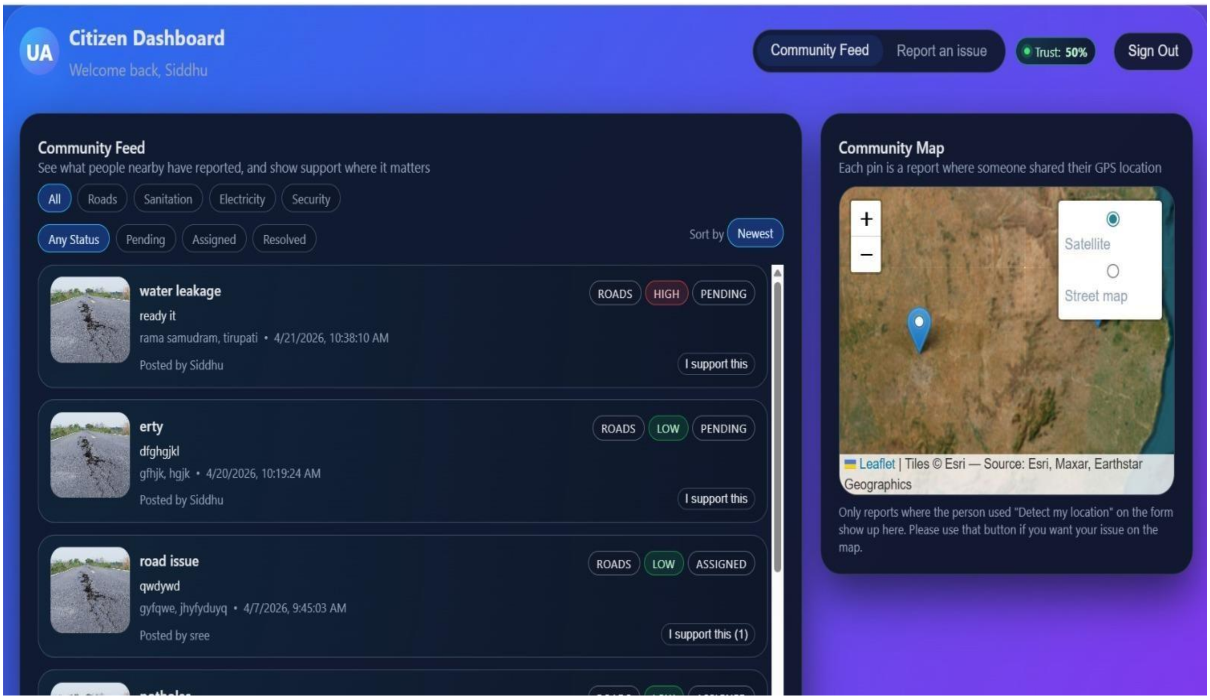
A.2 citizen login page



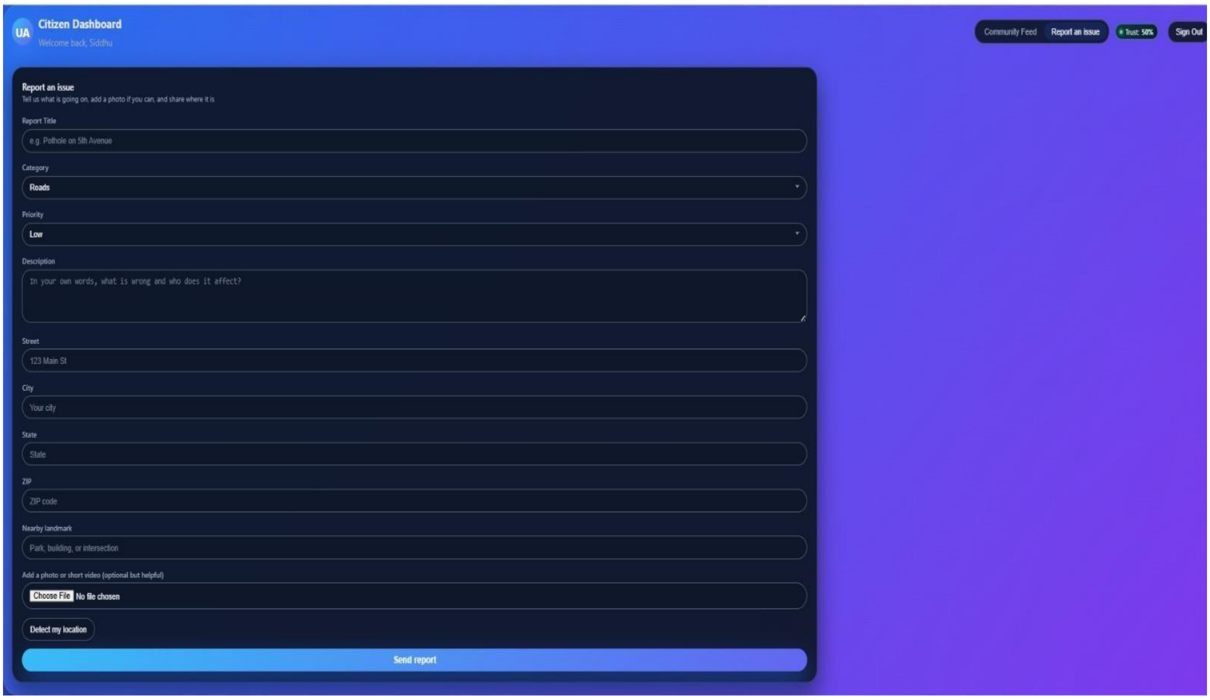
A.3 department staff login page



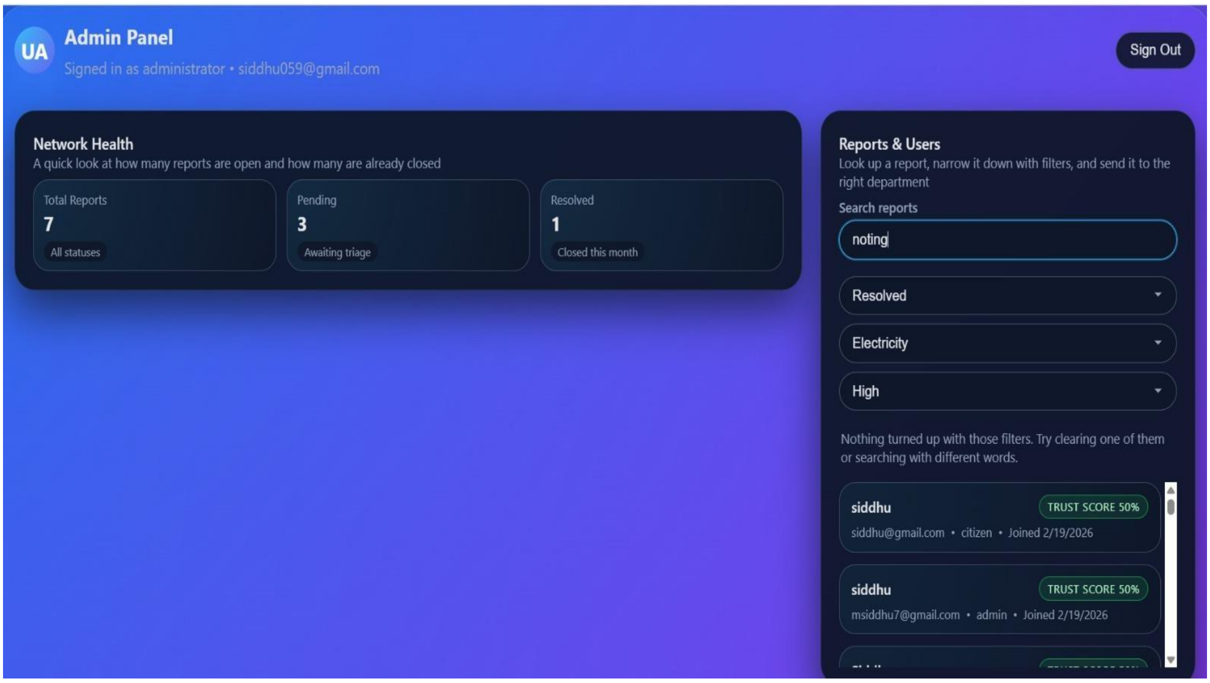
A.4 administrator login page



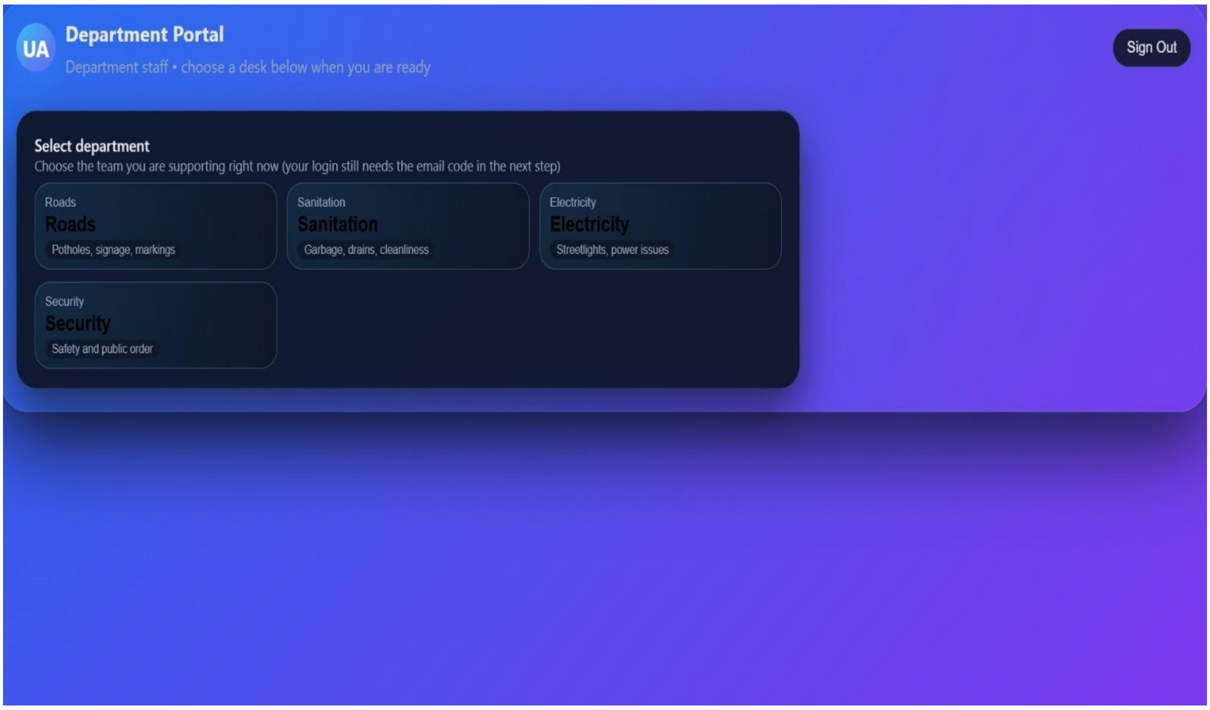
A.5 dash board



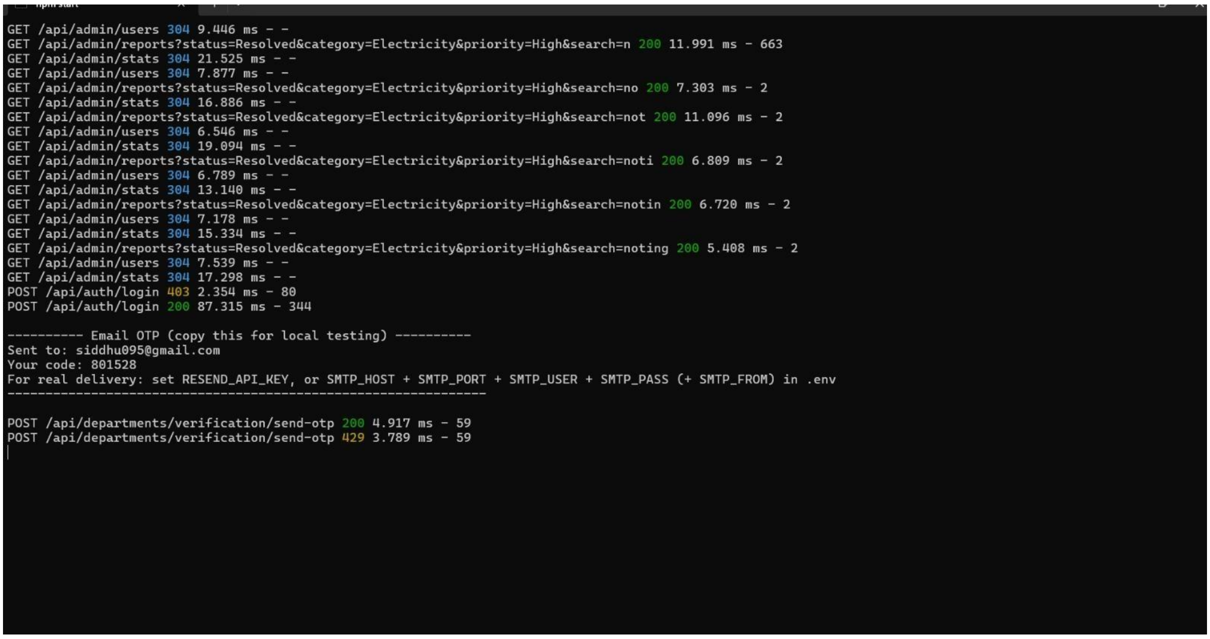
A.6 complaint page



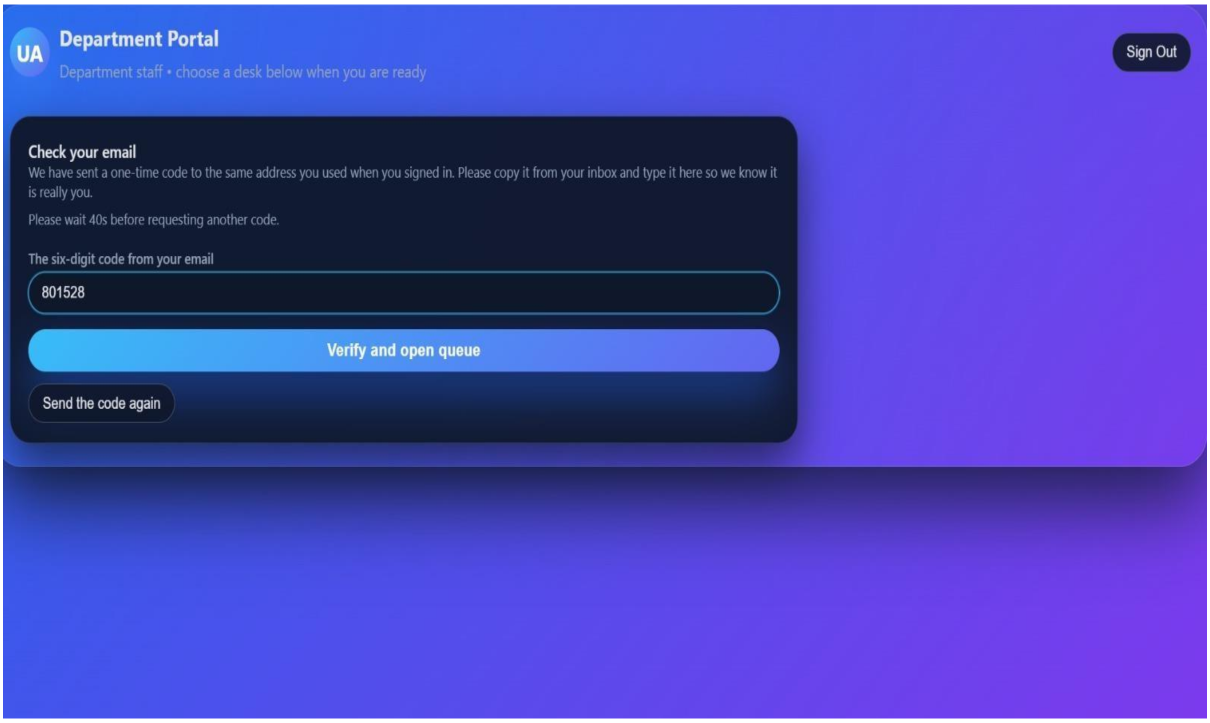
A.7admin panel



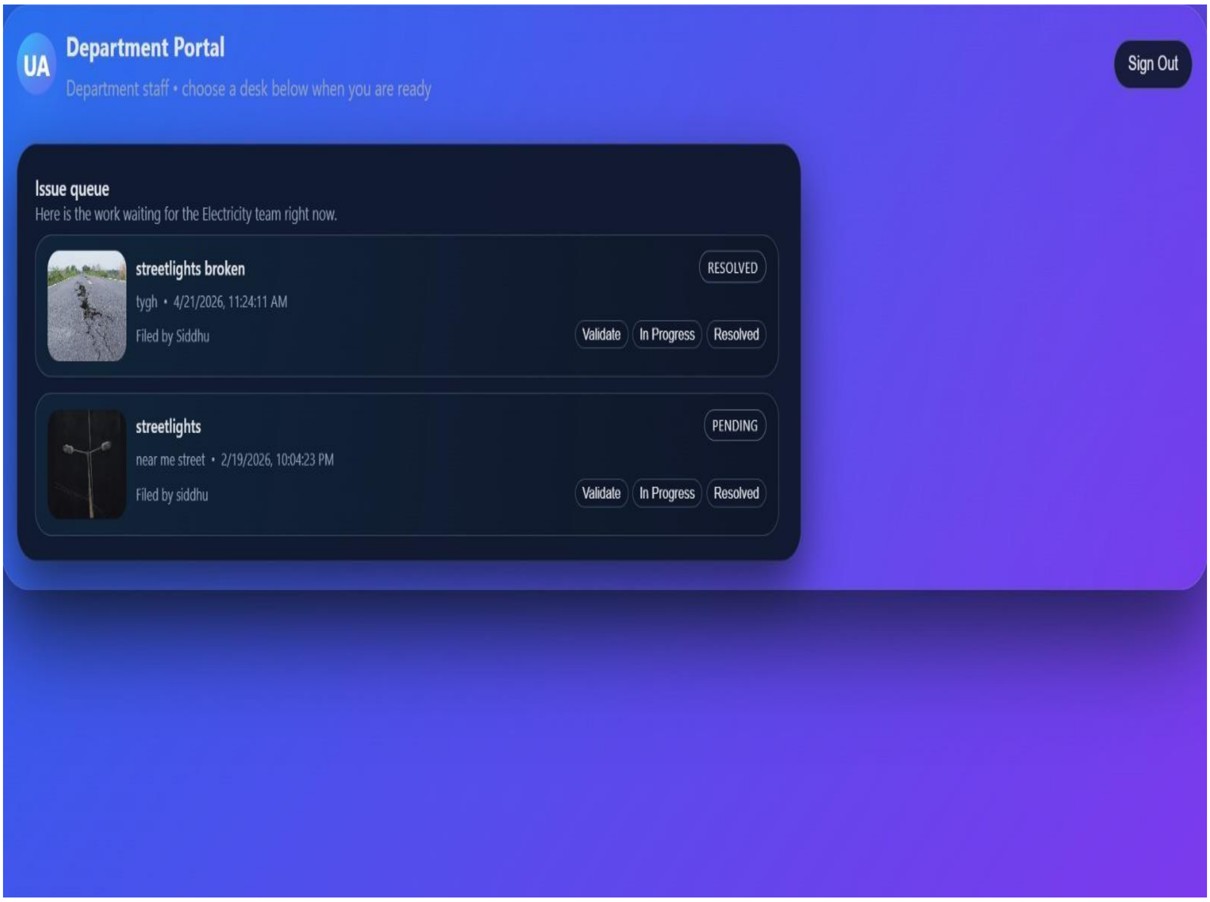
A.8 department panel



A.9

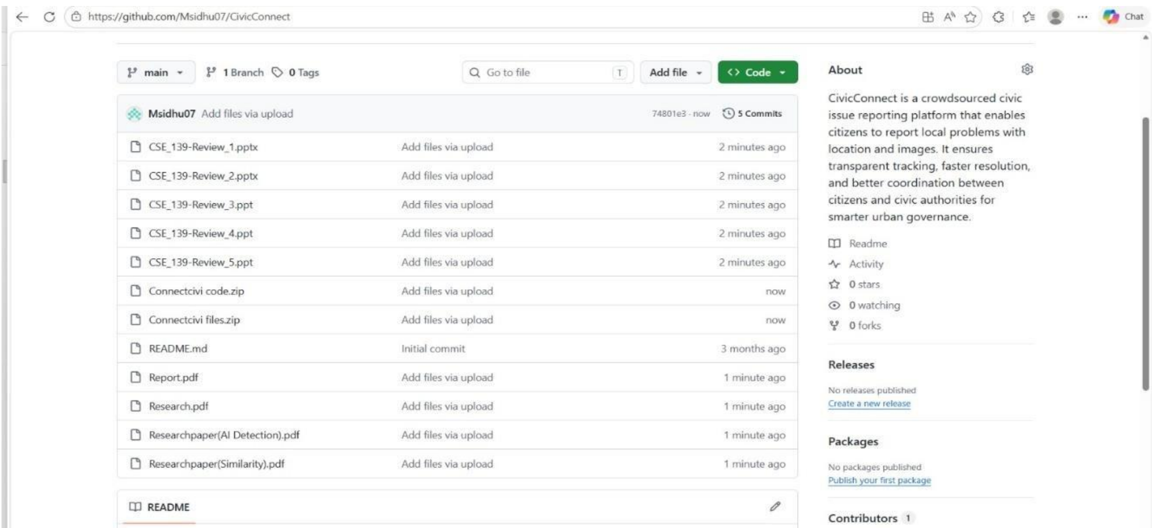


A.10 portal



A11 final output

APPENDIX B GIT HUB REPOSITORY



APPENDIX C SIMILARITY REPORT AND DETECTION REPORT

turnitin Page 2 of 8 - Integrity Overview Submission ID trn:oid:::1:3530845471

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **12 Not Cited or Quoted 4%**
  Matches with neither in-text citation nor quotation marks
- **1 Missing Quotations 0%**
  Matches that are still very similar to source material
- **0 Missing Citation 0%**
  Matches that have quotation marks, but no in-text citation
- **0 Cited and Quoted 0%**
  Matches with in-text citation present, but no quotation marks

Top Sources

- 4% Internet sources
- 1% Publications
- 0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

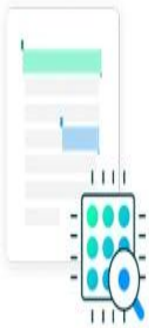
0 AI-generated only 0%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.